Harmanpreet Singh

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Software engineer with production full-stack experience at Lykan Technology (FastAPI, PostgreSQL, React, Docker, Kubernetes, AWS) and Stylabs Technologies (35+ features across web applications serving 50K+ users), and graduate depth in distributed systems, databases and networking (M.S. Computer Science, expected December 2026). Built HomeSense, a streaming platform on Kafka, PostgreSQL, FastAPI, React and Kubernetes, and SystemForge, a system-design simulator with a 549-test engine.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, PHP, Java, C, SQL

Backend and APIs: Node.js, Express, FastAPI, Flask, REST APIs, Socket.IO, OpenID Connect, API documentation

Data: PostgreSQL, MongoDB, MySQL, Redis, Kafka, Alembic migrations, indexing, query optimization, aggregation pipelines

Systems and infrastructure: TCP and UDP sockets, threads and locks, consistent hashing, DNS, AWS (S3, EC2, RDS), Docker, Kubernetes (EKS), Helm, GitHub Actions, Prometheus, Grafana

Frontend: React, Redux, Next.js, Vue 2/3, Nuxt, Expo (React Native), Tailwind CSS, HTML, CSS, Material UI

Testing and tools: Vitest, Playwright, Postman, Swagger, Jira, unit, integration and functional testing, debugging, Git, Linux

CERTIFICATIONS

Google Cloud: Manage Kubernetes in Google Cloud, Create and Manage Cloud SQL for PostgreSQL Instances. AMD AI Academy: Kubernetes on AMD Instinct GPUs - From Zero to AI Inference. DataCamp: AI Engineer for Developers Associate, GitHub Foundations. Databricks Academy: Databricks Accredited Generative AI Fundamentals. IBM SkillsBuild: Web Development Fundamentals, Data Fundamentals, Artificial Intelligence Fundamentals. Claude Academy: AI Fluency for Builders, Building with the Claude API, Claude 101, Claude Code 101, Introduction to Claude Cowork. OpenAI Academy: AI Foundations. EY and Microsoft: AI Skills Passport.

EXPERIENCE

Lykan Technology (now Lykan AI Studio), Full-stack developer July 2021 to March 2023

- Cut response times on high-traffic API endpoints with asynchronous FastAPI handling and tuned PostgreSQL query plans; cut AWS costs by shrinking Docker images and tuning Kubernetes limits; automated zero-downtime deployments with Git and Docker.
- Built high-throughput Python background workers on AWS services, and a shared React component library with strict TypeScript interfaces that reduced UI bugs and sped up feature delivery across teams.

Stylabs Technologies, Technical Intern, full-stack May 2023 to June 2024

- Shipped 35+ production features across 4 web applications (Manzil, Hireavilla, Profound Properties, Mainstreet) serving 50K+ users, with Vue 2/3, React, Next.js, Node.js, Express and MongoDB, including the chat support system and rental, sales and expense features.
- Built 40+ MongoDB aggregation pipelines over 50K+ records, cutting average query latency from 750 ms to 35 ms; cut marketplace page-load time from 4 s to 1 s with code splitting and lazy loading; wrote the Mainstreet API documentation, which improved user engagement by 27%.
- At Profound Properties, helped set up the foundation and built an AI-assisted seeding workflow for 5K records that saved 40 hours and helped deliver the project 12 days early.

Freelance: restaurant website, class attendance system, COVID-19 dashboard (HTML, CSS, JavaScript, React, PHP) January 2020 to June 2021

PROJECTS

HomeSense: smart-home platform (sole author; FastAPI, Kafka, PostgreSQL, React, Expo, Helm) June to October 2026

- Backend with OpenID Connect (PKCE) and optimistic locking; Kafka pipeline with event-time windowing, at-least-once delivery, idempotent writes and a dead-letter topic; live updates over server-sent events.
- Helm chart with 15 workloads and default-deny network policies; recovery verified by killing pods on a local three-node cluster. 27 architecture decision records, about 500 Python tests.

QueryDoctor: evidence-based PostgreSQL and MongoDB performance investigator Docker Compose

- FastAPI, Celery, PostgreSQL 16, HypoPG, MongoDB 7, Next.js. Tests index ideas with HypoPG and benchmarks approved ones in an isolated copy of the data; on the 100,000-order sample one index cut a median from 8 ms to 0.32 ms.

SystemForge: interactive system-design laboratory Live site

- Next.js 16 and TypeScript. Deterministic simulator behind 40 system labs, an architecture canvas and a sandboxed code playground; real Raft, OT/CRDT and order-book implementations; 549 tests.

DISTRIBUTED SYSTEMS AND DATABASE PROJECTS

Distributed key-value naming service on a consistent-hashing ring, multithreaded FTP, persistent multicast (Java, TCP); DNS forwarder with DNS-over-HTTPS (Python); HTTPS downloader (C, OpenSSL).

Relational algebra engine with hash indexing and predicate-aware query plans (Java); property management system, team of three (Nuxt 3, Express, PostgreSQL).

RESEARCH AND PUBLICATIONS

FACET, a smart-home activity recognition paper under review at a conference (joint work). M.S. thesis on multimodal self-supervised activity recognition (CASAS). Resume Ranking System Using Machine Learning and NLP, IJSREM, May 2023.

EDUCATION

M.S. Computer Science (thesis track), University of Georgia, Athens, GA August 2024 to present, expected December 2026

GPA 3.57/4.0. Coursework: Algorithms, Distributed Computing, Database Management, Computer Networks, ML in IoT

B.E. Computer Engineering, Savitribai Phule Pune University August 2019 to June 2023

GPA 8.7/10, First Class with Distinction